

Data Matrix and Dissimilarity Matrix

- Data matrix

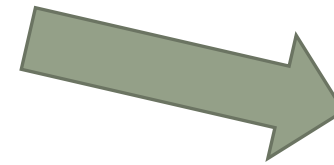
- A data matrix of n data points with l dimensions



$$D = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1l} \\ x_{21} & x_{22} & \dots & x_{2l} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nl} \end{pmatrix}$$

- Dissimilarity (distance) matrix

- n data points, but registers only the distance $d(i, j)$ (typically metric)
 - Usually symmetric, thus a triangular matrix
 - **Distance functions** are usually different for real, boolean, categorical, ordinal, ratio, and vector variables
 - Weights can be associated with different variables based on applications and data semantics



$$\begin{pmatrix} 0 & & & \\ d(2,1) & 0 & & \\ \vdots & \vdots & \ddots & \\ d(n,1) & d(n,2) & \dots & 0 \end{pmatrix}$$

Standardizing Numeric Data

□ Z-score: $z = \frac{x - \mu}{\sigma}$

□ X: raw score to be standardized, μ : mean of the population, σ : standard deviation

□ the distance between the raw score and the population mean in units of the standard deviation

□ negative when the raw score is below the mean, “+” when above

□ An alternative way: Calculate the mean absolute deviation

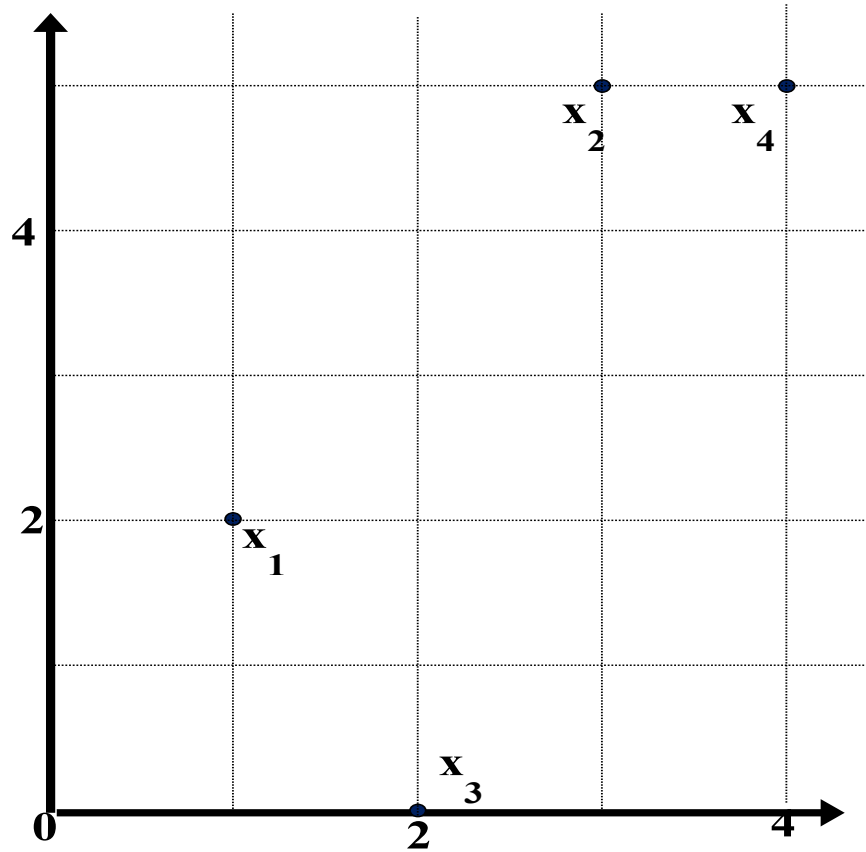
$$s_f = \frac{1}{n} (|x_{1f} - m_f| + |x_{2f} - m_f| + \dots + |x_{nf} - m_f|)$$

where $m_f = \frac{1}{n} (x_{1f} + x_{2f} + \dots + x_{nf})$.

□ standardized measure (z-score): $z_{if} = \frac{x_{if} - m_f}{s_f}$

□ Using mean absolute deviation is more robust than using standard deviation

Example: Data Matrix and Dissimilarity Matrix



Data Matrix

point	attribute1	attribute2
$x1$	1	2
$x2$	3	5
$x3$	2	0
$x4$	4	5

Dissimilarity Matrix (by **Euclidean Distance**)

	$x1$	$x2$	$x3$	$x4$
$x1$	0			
$x2$	3.61	0		
$x3$	2.24	5.1	0	
$x4$	4.24	1	5.39	0